



GLASSBRAIN

GlassBrain LLC.

Founded by Cade Harger

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Executive Summary

GlassBrain LLC.

GlassBrain was created to break down the barriers that prevent people from learning about neural networks. We are creating a desktop application that allows users to begin learning neural network concepts in a way that is more intuitive, visual, and hands-on than any existing method. We believe the key to managing the great power and risks that come with Artificial Intelligence is to make concepts accessible and easy for anyone to grasp.

Overview of Business

GlassBrain is a computer desktop application that uses a first-ever drag-and-drop building block interface that lets users create and experiment with [neural network](#) models. Integrated into the app on top of this, GlassBrain has learning modules (resembling mini-courses) that interactively cover different neural network concepts starting with the most basic and progressively building upon one another. GlassBrain's unique drag-and-drop interface makes content easier to understand, boosting the learning process within the modules and creating a better learning experience than any other existing option. GlassBrain seeks to be the go-to entry point for anyone considering learning about machine learning.

Currently, GlassBrain is in development, with the drag-and-drop interface nearing completion. GlassBrain plans to have the desktop app finished and learning modules developed to be ready for a release at the beginning of summer 2024.

GlassBrain offers a Free Version that includes the desktop app with the neural network creation interface as well as access to three learning modules of the user's choosing. After using the three free modules, users will be able to purchase more on a per-module basis for \$2-10 depending on the module. GlassBrain also offers a Premium Version that includes access to all modules and a few additional features.

Market Overview and Strategy

GlassBrain estimates the Introductory machine learning education market to be roughly equivalent to \$400 million, with GlassBrain positioned to achieve a valuation of \$6 million in 3 years. GlassBrain's products, distributed on its website, are for any *individual* who wants to learn machine learning concepts, whether out of curiosity or career interest. Later

on, GlassBrain plans to scale by negotiating contracts with educational institutions (schools, bootcamps, online courses, etc.) for integrated use. GlassBrain's low price and one-of-a-kind form of representing concepts will help it stand out from its competitors, which primarily are online courses and decentralized learning resources available on the internet. GlassBrain plans to jumpstart a longer-term product-led growth strategy by utilizing free marketing in machine learning forums as well as spending in marketing channels where people search to learn machine learning topics such as YouTube and Reddit.

Operations

GlassBrain will distribute its desktop app through its website. Premium Version and Individual module sales will also be on the website, revealing product keys that can be used to unlock the modules in-app. The nature of GlassBrain being a desktop app makes for very low operational complexity and near-zero input costs post-creation. These low input costs will allow GlassBrain to reach profitability shortly after release.

Management

GlassBrain LLC is currently solely founded and run by Cade Harger, a Computer Science major at Trinity University. Cade has 8 years of coding experience in 10 different programming languages and is self, school, and online course-taught in machine learning since January 2021. Cade has released software before and has taught a summer camp teaching children how to code using a drag-and-drop interface similar to GlassBrain. Once GlassBrain releases, GlassBrain plans to hire up to two new employees experienced in sales and marketing to assist with the business expansion.

Funding

GlassBrain is seeking to receive \$25,000 in non-dilutive funding from winning the Stumberg Entrepreneurship Competition Final Round. It will use this funding to help create reputable educational content, scale the business, and bring on employees.

Idea Genesis

In January of 2021, When I heard stories of an AI program being able to generate faces of people that weren't real, I was intrigued. I had been programming for many years, and yet I couldn't imagine any traditional algorithm that could come close to doing something like that. I began to spend my spare time looking into AI and began falling in love.

But the learning process was very frustrating, and it felt like nothing was designed for beginners. I had just spent the summer teaching kids to code using Scratch, a visual building-block playground way to program, and had seen how effective it was in teaching them coding concepts. I thought, surely something like that exists for AI. I immediately imagined what it would be like, and began searching for it so I could start using it to help me learn because I knew it would make my life so much easier. But it didn't exist.

It will soon though.

- Cade Harger, Founder of GlassBrain

Products

GlassBrain Free Edition

– 75% complete, expected completion December 2023 –

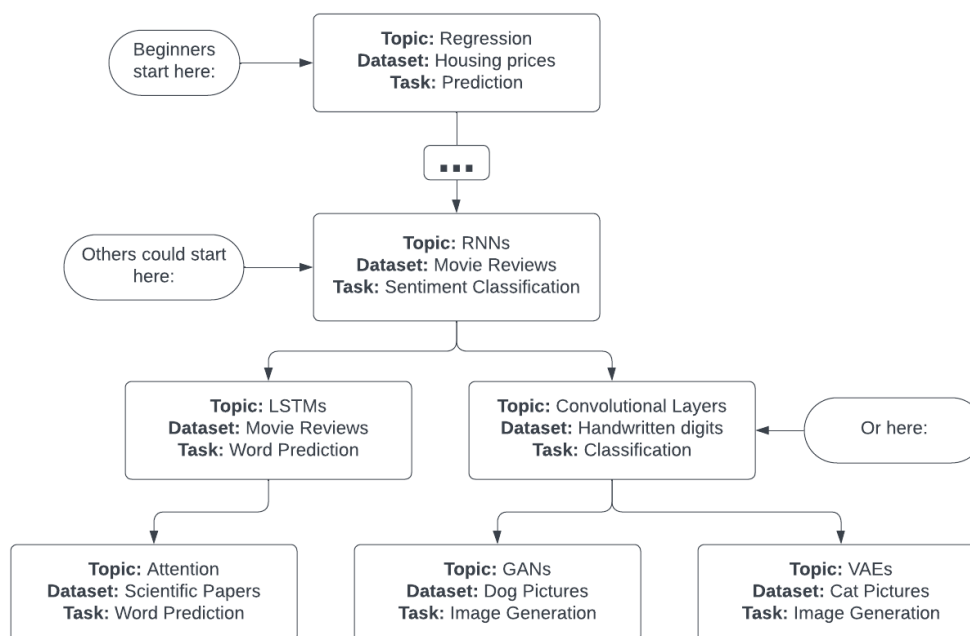
GlassBrain Free Edition is a free desktop app with a drag-and-drop interface to graphically build neural networks. Free users also get access to any 3 integrated learning modules of the user's choosing free of charge:

GlassBrain Learning Modules

– Expected MVP (~10 created) completion Summer 2024 –

GlassBrain Learning Modules are interactive lessons built into the desktop app guiding the user through the creation of neural network projects. Modules are a \$2-10 one-time purchase depending on the quantity of learning material.

An example module might teach a user about Convolutional Neural Networks (CNNs). Almost like building a Lego set, they would be guided by the app to drag each block in to build a CNN from scratch. Each block's purpose will be explained. Then the user would prepare a dataset (loaded from an external website) of pictures of cats and dogs. They would train the classifier they built to classify the pictures (This image is a cat, this is a dog). Learning Modules are an effective way of structuring content because they allow people who may have some prior experience to begin learning at their respective start. Here is a hypothetical of what a suggested continuation of modules a user would look like:



While Learning Modules will be created in-house, they will be reviewed and critiqued by experts before release.

GlassBrain Premium

– Expected MVP completion Summer 2024 –

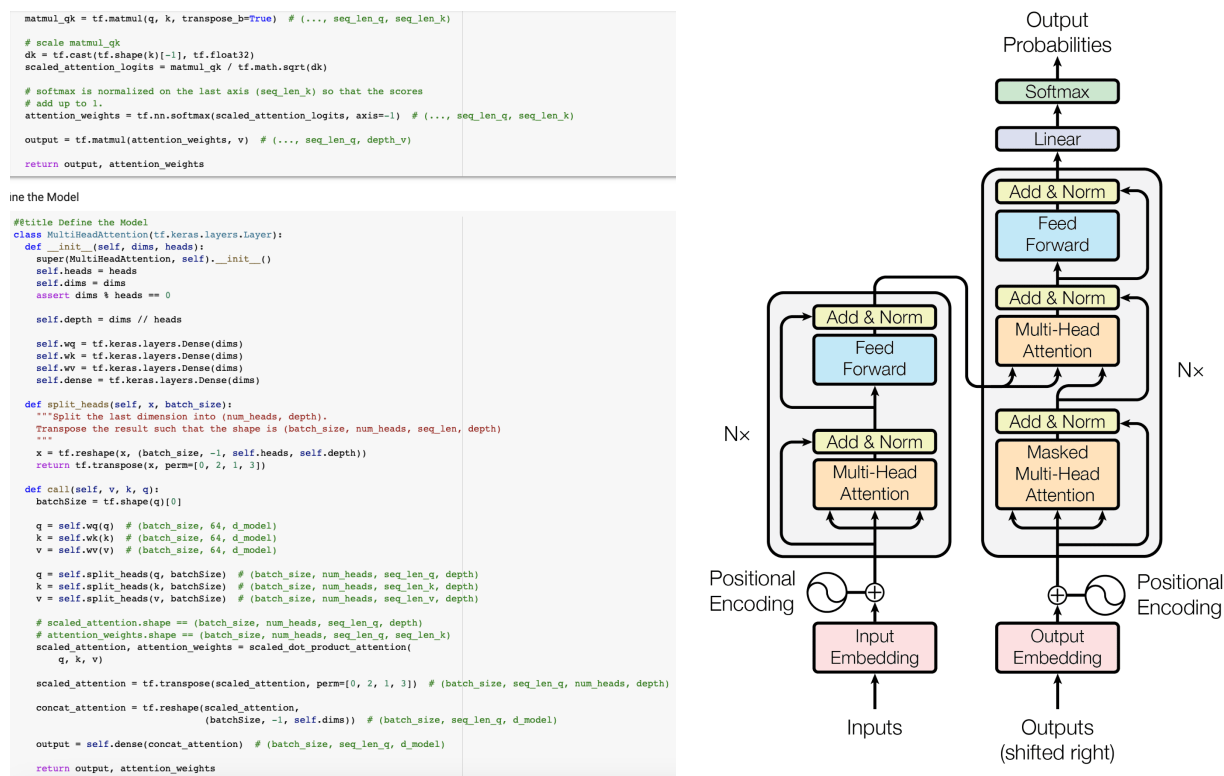
GlassBrain Premium will be an add-on to the GlassBrain Free Version. Premium is a \$50 one-time purchase with the primary benefit of being able to access all GlassBrain Learning Modules free of additional charge. GlassBrain Premium will also include a few additional features:

- Advanced explanations of each building block and what it does
- Ability to export projects (.h5 or Pytorch code)
- Certificate upon completion of all modules

What makes GlassBrain Uniquely Valuable

Fair warning, while written in the most universally understandable way possible, this gets fairly technical. It explains why the drag-and-drop interface that GlassBrain devised is better than existing alternatives.

GlassBrain teaches about neural networks and how to solve problems with them. The issue with the learning process for neural networks is that all learning resources use ways of representing the concepts that fall into two categories: Flowcharts/diagrams and Code:



An Example of code and an example of a neural network diagram. All of the code on the left represents the same concept as one box (Multi-Head Attention) in the diagram.

Code is used as a representation of concepts when learning because it is the only effective way to implement neural networks (right now). Code is designed to be efficient for people who already understand concepts, not for learners, but because it is the only way to implement concepts, the learning process is dependent on it. This creates many issues:

- Needing to know how to code creates an unnecessary entry barrier.
- Code can be intimidating to beginners or people who haven't worked with it before.
- Code fails to capture the bigger picture.
- Lines of code are often black boxes as they assume you are already knowledgeable of concepts. They often reference hundreds of lines of code from an external source, and the only way to figure out how they work is by turning to other resources.

To alleviate these issues, over the years educators began using flowcharts/visual diagrams in conjunction with code. One flowchart can communicate hundreds of lines of code in seconds. Flowcharts also help learners see patterns in the way networks are built that would otherwise be hard to grasp from reading dozens of lines of code. They fill in the gap that code leaves in understanding. We could never only use flowchart pictures, however, because unlike code, they don't work. You can't take a picture of a neural network diagram and run it. Like code though, flowcharts are also black boxes because they are designed to just give a bigger picture. They leave out small details, but those details are valuable information to fully understand what is happening.

But bouncing between these two representations creates a harder, more decentralized learning process. Learners find themselves jumping back and forth between learning resources because they don't have a proper balance of the two types of representation. They also may do this to open the black box and figure out how something works. For example, an online course someone is taking might use only code, so the user seeks out a YouTube video on the topic to get a more visual, big-picture understanding. But this YouTube video uses a different notation than their course. They repeat this process a few times to help bridge the definitions they saw in one resource to the ones they read in another. A GlassBrain survey respondent described it as "...learning piecemeal from 50 different websites that use 50 different notations".

Even courses, which exist to centralize learning processes, suffer from this issue because they are still confined to only coding and visual diagrams as representations. One slide in a presentation might have code, another might have a formal written definition, and then another may have a diagram, all to describe the same concept. If learners had one quality form of representing concepts, this could be one slide.

The Solution

GlassBrain's desktop app creates an interface that combines the representations of code and flowcharts into one, resolving the aforementioned issues. Users can create flowcharts that actually function by dragging and dropping building blocks. There are no black boxes because users can double-click on any node in a flowchart to open it up and see its subcomponents. Additionally, clicking on a node will display an explanation of what the block does and why it exists. Users don't need to know how to code to get started learning with GlassBrain because code isn't used as a learning resource. Users can recognize patterns in the networks they build much more easily because their networks are visual.

GlassBrain has created a representation form that is complete and optimized for people learning neural networks. **There is no existing form of representation that does this until now.** Every other method of learning uses static flowcharts and code. That is what sets GlassBrain apart from every other method of learning neural network concepts.

Intellectual Property Protection

GlassBrain has taken and will take several measures to ensure the protection of its intellectual property:

- Educational Modules and the Desktop app will include copyright notices.
- Terms of use will clearly outline what is acceptable and will be looked over by experts.
- GlassBrain has filed for a trademark for its name.
- GlassBrain will limit direct marketing until there is a product to buy.

Patenting is an expensive and time-consuming process, so this will be something that GlassBrain will only seek to do once significant profitability has been achieved.

Market and Strategy

Industry Description

GlassBrain is in the industry of Introductory Machine Learning Education, described as all businesses that offer education for the initial concepts in the field of machine learning. GlassBrain's Premium version and Learning Modules serve as entry points into learning about machine learning. GlassBrain will continue to expand within the industry through integration with other places people go to begin learning ML (schools, bootcamps, online courses).

Industry Size Analysis:

Based on the following data as well as founder experience, GlassBrain estimates the Total Addressable Market (TAM) for Introductory Machine Learning Education to be worth \$350 Million:

- The global AI industry is currently valued at [\\$207 Billion](#) and is expected to grow rapidly.
- “The US edtech market size was valued at \$58.1 billion in 2021 and is expected to achieve a CAGR of more than 20% during 2021-2026” - [globaldata.com](#)
- The [Coursera](#) Top 5 Machine Learning courses all cost \$50/month and have a total of 1,883,465 already enrolled for a total of \$94,173,250 in sales.
- The [Udemy](#) Top 5 Machine Learning courses have prices ranging from \$20-110 with a total of 1,870,747 students for a total of \$195,247,982.50 in sales.
- The [top YouTube video](#) for the search “learn neural networks” is 5 years old and has 14.4 million views.

The primary barriers to GlassBrain reaching this entire market are:

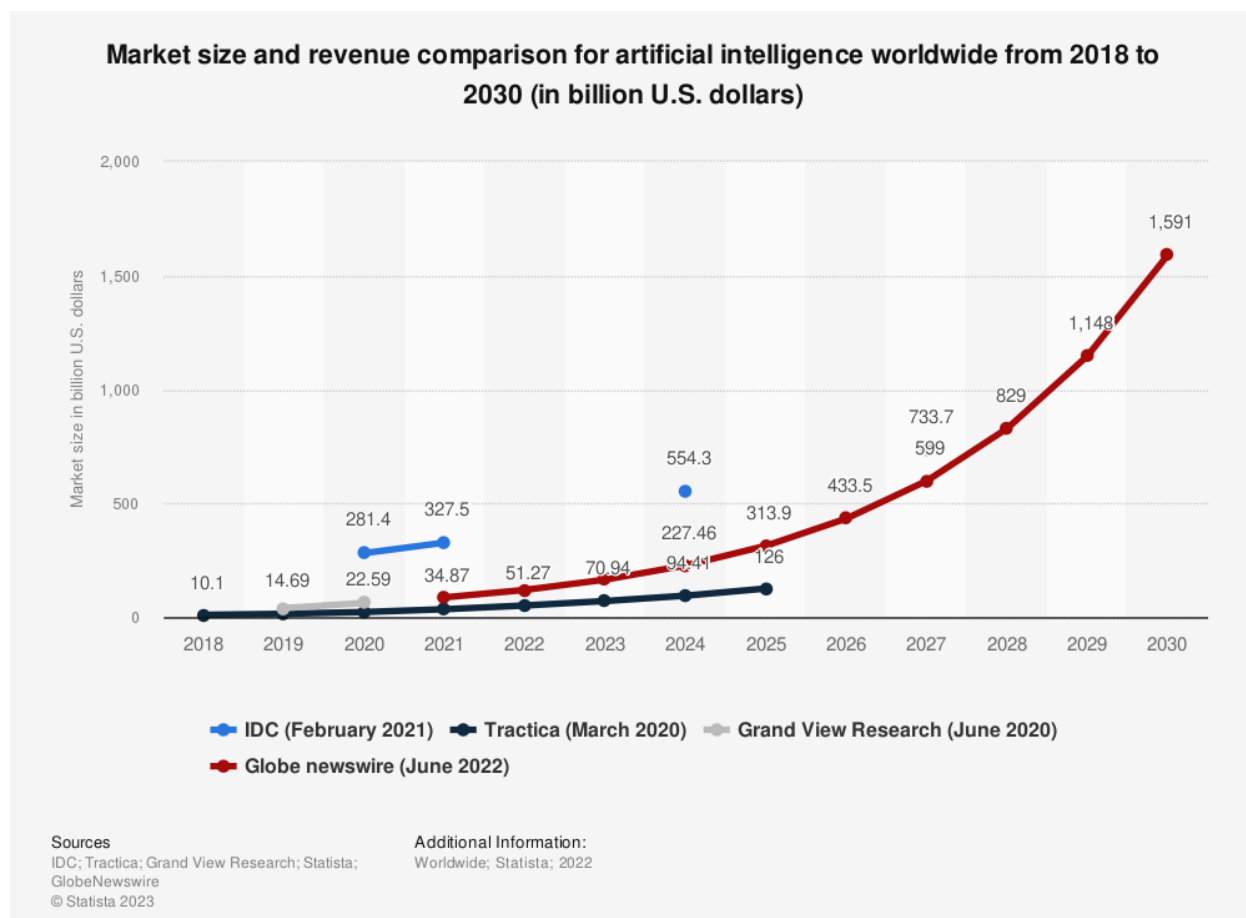
- GlassBrain is a paid resource and not monetized through advertising, thus out of some budgets.
- GlassBrain requires a desktop computer with space to download with average to above-average specs.
- GlassBrain will initially only be available in English.

With these barriers considered with the TAM data, a reasonable estimate for GlassBrain's Serviceable Addressable Market (SAM) is \$100 million.

As discussed later on in this section, GlassBrain is extremely differentiating on an axis that no other learning method differentiates itself on. Given this, a conservative estimate of the Serviceable Obtainable Market (SOM) that GlassBrain would control 3 years from now is \$5 million.

Industry Trends

The industry is expanding rapidly: With recent breakthroughs in technology, the AI industry is exploding. The expected growth rate over the next 10 years is extremely high:



Education is behind: Trinity University only offered one course (non-seminar) in the Fall 2023 semester focusing on machine learning despite high student interest.

Education is trying to catch up and willing to spend to do so: Trinity University recently hired [Dr. Eva Tuba](#), a machine learning professor with over 3,000 citations.

All of these trends support the extreme growth of the demand for solutions like GlassBrain over time.

Individual Demographic

Any individual interested in getting started learning machine learning for any reason.

Needs:

- To learn machine learning while minimizing time spent, money spent, and difficulty. People tend to learn machine learning for two reasons:
- Satisfy curiosity: With all of the headlines AI has been making recently, people are more interested than ever.
- Change/boost career: There is a lot of money and opportunities in the AI space right now. GlassBrain is the perfect starting point for someone looking to chase these opportunities.

GlassBrain resolves these needs not just by giving them an opportunity to learn. It makes learning more intuitive with its one-of-a-kind building block representation, reducing time spent and difficulty. You can also only pay for what you want to learn with the module-based learning model, reducing money spent.

Behaviors:

- Individuals entering the learning process often see the vast amount of decentralized learning resources available on the internet and will turn to peers or forums to find a recommendation on how to navigate them.
- A large portion of these individuals will be computer science undergraduate students. Many computer science students create personal projects such as creating their own website or game in order to gain experience and demonstrate knowledge to employers. GlassBrain would be a low-effort and entertaining way to accomplish this.
- Learning through methods like GlassBrain is extremely common in computer science, especially when compared to other professions. When asked “How do you learn to code? Select all that apply.”, [Stack Overflow’s 2023 Annual Developer Survey](#) respondents answered:



Characteristics:

- Likely ages 16-35 with some previous interest/experience in a science-related field
- Prerequisite knowledge:
 - Coding experience. All existing solutions require this *except GlassBrain*.
 - High school-level math (Algebra, Statistics)

Early Adopters:

As this is an entry point for people getting started in the field of machine learning, people looking for solutions don't have much past experience to go off when choosing a learning method. Because of this, the early adopters of GlassBrain will be roughly the same demographic as the regular individual, except they might look further for a solution to their pain points and desire a more visual, hands-on learning process. Additionally, GlassBrain will primarily target the undergraduate computer science student subsector of this demographic first, as that is the group of people GlassBrain best understands.

Why Individuals First, and not Schools?

Selling to a market of individuals first creates a foot-in-the-door to opportunities to work with educational institutions. This allows GlassBrain to build its reputation as having a proven ability to help people learn, something that educational institutions expect before making deals. Bugs and issues are also more acceptable on an individual basis, as they only prevent one person from learning instead of a whole class of people where the institution's reputation is on the line. This will give time for GlassBrain to resolve these problems.

Educational Institution Demographic

Because GlassBrain solves a problem that is fundamental to all existing ways of teaching neural networks, it can be integrated into almost any existing way people learn. Once GlassBrain has built a positive reputation by selling to individuals, it will expand to negotiating contracts with educational institutions (**high schools, undergraduate schools, graduate schools, bootcamps, and online courses**) where GlassBrain will be directly integrated into lessons. This could be where classes complete learning modules together, or where teachers can create their own modules, or where GlassBrain provides lesson plans. There are many possibilities.

Needs:

- To catch their institution up to the rapidly advancing field of AI
- Hiring an AI expert could be very expensive, but a one-stop solution to resolve the lack of AI knowledge would be optimal
- Only willing to spend money on things with proven track records to help people learn.

Characteristics:

- Budgets in schools are usually allocated at a specific time in the year and not done again until the next year. Finding this period for each client is crucial.
- While educators often have some influence, budgets are usually allocated by decision-makers who are higher up in the chain of command within institutions. For Trinity Computer Science, the process only needs the approval of the department head, which is one of the professors, unless it is very expensive (more than GlassBrain would be). In that case, it would go to the department head, then the dean, then Academic Affairs before finally being approved.
- Institutions with an emphasis on project-based learning or visual learning
- Budgets range from \$1,000 - 20,000

Behaviors:

- Decision makers can be reached through connections, teacher/subordinate influence, or cold channels (email, call, etc.).

Early Adopters:

Marketing channels for decision-makers who are interested in machine learning are much more narrow than those for individuals. Early adopters will likely just be the first institutions to agree to meet after extensive cold-reach strategy or through connections.

Competition and Differentiation

GlassBrain's competition consists of any complete learning method that serves as an entry point into the field of machine learning.

Online courses: These can either take place independently or more commonly through course platforms, with the largest two being [Coursera](#) and [Udemy](#). Many different paid and free course options exist, with prices ranging from [several thousand](#) to many [free options](#). These options differentiate themselves based on price point, course length, depth, support, and prestige of the institution/educator.

With its Learning Module system, GlassBrain allows you to control how long/deep your learning goes. While GlassBrain offers minimal support, its price point is on the lower end of comparable methods. By having Learning Modules critiqued by experts, it attains a reasonable level of prestige/quality for its price point.

Free online resources: These resources are usually free and monetized through advertisements. They are different from courses by the fact that they are decentralized from a resource standpoint. They come in the form of one-off articles and videos often on YouTube, Google, or Blogs. These resources differentiate themselves to the consumer primarily through how accessible they are, usually through search engine optimization (SEO). While each resource is different in the way they teach/represent the concepts, the consumer isn't aware of these differences normally until they have already begun learning.

GlassBrain will compete with these resources by advertising in channels that users search for them, specifically mentioning pain points that arise within the decentralized learning process.

In-Person School courses/Bootcamps: These are almost identical to online courses except they tend to be higher in quality, support, and price point. Differentiation is less important because students take what options are available to them. Many schools offer none to very few ML courses and are likely outdated given the rapid technological expansion recently.

GlassBrain will also seek to avoid competition eventually by integrating into each of these learning methods.

Differentiation

GlassBrain's drag-and-drop interface [solves a problem](#) that is fundamental to all existing ways of learning neural networks: The lack of a good concept representation form. This representation allows learners to have an easier time understanding concepts, making it

more appealing in that aspect than any existing learning method listed above. Additionally, this representation removes the entry barrier that is the need for people to know coding, which no other learning method does. This essentially creates a monopoly for the small subset of people who don't code but are interested in learning neural networks at a considerable depth.

This representation also helps it visually stand out in a public or advertising setting. Anyone who has spent more than a week learning about neural networks will be able to recognize within 10 seconds of looking at GlassBrain that it uses functioning diagrams as a concept representation, something they will have never seen before. They should be able to conclude that this will result in an easier learning process.

GlassBrain's module-based learning system also differentiates itself from competitors because it allows people to start without commitment, not just from the start of a course but from any point because the user can pick any 3 modules for free. This system is appealing to people who have started their learning elsewhere and have frustrations and want to try another way of learning with no commitment. This makes it more cost-effective than switching to another course, as they would be paying for some content they already have learned if they purchase another entire course. This way, they only pay for the modules they need.

Market Acceptance

There are many indicators that GlassBrain is an idea that the market wants:

The precedent for GlassBrain exists. [Scratch](#) is a graphical drag-and-drop interface that instead of building neural networks like GlassBrain, builds computer programs. Its visual interface is designed to teach children or beginners the ideas behind programming without dealing with the stress of code. This is exactly what GlassBrain does: in the long run, GlassBrain probably isn't a replacement for coding. But it takes away the pain points from the learning process and teaches the underlying intuition. This way, if you do need to learn to code neural networks one day, it will be very easy. Having this precedent makes GlassBrain and its benefits easily recognizable to people who have used Scratch before. Scratch has over 100 million registered users worldwide. When talking to people about GlassBrain during the summer accelerator, it was often the case that telling someone GlassBrain is like "Scratch but for AI" was enough to communicate what it does and convince them it is a good idea.

Discussion on learning methods is widespread. There is a [Subreddit](#) with 342K members dedicated to discussing learning methods and helping people learn machine learning.

People have recognized the need for GlassBrain and built similar things, but none of them have had the same scope as GlassBrain or created the same drag-and-drop representation:

- [Tensorflow Playground](#) (Free): This is like a very simplified version of GlassBrain. You can create neural networks and change their architecture visually, but only up to 48 neurons. You also are restricted to a strict layer structure. This is good for demoing what a singular neuron does, but not much more.
- [MATLAB Deep Learning Toolbox](#) (\$29 + MATLAB (\$99) if student, up to \$3,800 otherwise): This allows you to create functioning neural network diagrams and modify architecture but only at a base level. You cannot modify the internal structure of blocks or create your own dynamic blocks, so it is not a ‘complete’ representation and thus not useful for learning. This is also outdated.
- [TensorBoard](#) (Free): Visualize neural networks. Not designed for learning and you must create the networks with code first in order to visualize them.
- A free [neural network video series](#) by 3Blue1Brown, a math visualization YouTube channel, is widely known and used in the ML community for learning about neural networks because of how well it integrates visuals and neural network concepts. While this alone is a validation as a sign of needing visual resources, the series has a significant dropoff with the first video having 14.4 million views and the last (4th) having only 2.4 million views. If such a widely appreciated resource has this big of a dropoff, there is a problem.

Customer Survey Highlights

After recording conversation notes I had and conducting [surveys](#), a few points stood out reinforcing the issues GlassBrain was founded to solve:

- The majority of people interested in AI but hadn’t learned yet cited the lack of a clear entry point as a primary barrier. This is likely tied to the lack of differentiation in the field of learning methods.
- “[Scientific] papers had partial information to what actually is being used” - A learning resource that is designed to explain an idea isn’t sufficient because of poor learning representations.

- “Poor diagrams of neural networks that don't explain the underlying operations being performed” - Diagrams are useful resources until you don't know what one of the blocks does. GlassBrain solves this by allowing you to double-click on a diagram block and see its subcomponents.
- “Tensorflow [Neural Network Coding Framework] didn't have enough freedom for my liking, so I started developing models only from scratch” - Similar to diagrams, lines of code exist for efficiency so they are short and abstract, but this is not ideal for someone trying to experiment with them to learn.
- “[There] is not a bridge from the basic stuff on videos to the really in-depth explanation in papers.” - Not only does this demonstrate that this person had to use multiple learning resources (videos and papers), which is time-consuming in and of itself, but they also cited the discontinuity between resources, which leads to further confusion for learners.
- “...learning piecemeal from 50 different websites that use 50 different notations” - This also highlights the decentralized learning process that results from no resource being sufficient on its own. GlassBrain is one app with one notation that has all the information a learner needs.

Market Penetration Plan

Given how differentiating GlassBrain is from its competitors and how much discussion around machine learning there is, a lower-cost, more product-led growth strategy will be effective. To jumpstart this strategy and get a word of mouth started, GlassBrain will employ the following tactics:

- Free advertising by posting and fostering a community on targeted forums and social media:
 - Machine learning subreddits (listed have a total of 4.6 million total members): [r/learnmachinelearning](#), [r/machinelearning](#), [r/neuralnetworks](#), [r/artificialintelligence](#), [r/deeplearning](#), [r/artificial](#), [r/openai](#), and more
 - StackExchange, Instagram, Facebook, Twitter, LinkedIn posts
- Exploring paid marketing through channels that people search to find resources to learn machine learning:
 - Paid social media ads

- Sponsoring YouTube videos related to neural networks, especially videos that involve more visualization
- Search Engine Pay Per Click (PPC) ads on learning keywords
- Search engine optimization geared towards learning machine learning, with an emphasis on visualization.
- Affiliate marketing: Giving referral bonuses to partners and potentially receiving them as well with similar products that focus on online education for similar topics but aren't direct competitors.

Once downloads and sales begin coming in, GlassBrain plans to refine its product using the following strategies:

- **Landing page optimization:** GlassBrain intends to use A/B testing to optimize the landing page. Potentially freelance sales work would be useful.
- **Focus Group(s):** GlassBrain also intends to hold a focus group at some point, where users are paid to learn machine learning using GlassBrain while being observed and discussing their experience to increase product quality.
- **Feedback Collection:** User feedback will be collected after every module is completed. This along with bug reports and feature usage will be analyzed to adjust features to further improve product quality.
- **Parent's Guide:** GlassBrain will create a document titled "Parent's Guide" that is linked on the landing page and will contain information necessary to give parents the confidence to purchase GlassBrain.

Launch Plan

Before the launch, GlassBrain is considering releasing its free version once it is finished, likely in January of 2024. This could help build hype around the Premium/module releases. To create time to iron out bugs and other issues, GlassBrain plans to soft launch its Premium version/modules once complete (expected in the Summer of 2024). This ensures marketing efforts will not be wasted and minimizes any reputational damage done by unforeseen issues during launch. Reputation is especially important for such a differentiating product using a product-led growth strategy.

Branding

Values: Accessibility, Creativity, Experimentation, and Intuition

GlassBrain believes its interface and Learning Modules must evoke its values to be successful so they are designed to reflect them. Existing learning methods have very little branding and don't feel exciting or empowering. Neural network concepts are fascinating to many and the things that can be made with them are incredible, so there's no reason it should be this bland.

GlassBrain's dark-mode aesthetic with the occasional vibrant purple and blue gradient gives it a futuristic feeling – and rightfully so, users are learning about cutting-edge technology. Naming schemes for products should feel empowering and inspire curiosity. For example, GlassBrain Premium might be called GlassBrain Genius, or something similar during release. Learning Modules will feel friendly and questions learners may have will be anticipated and answered, fostering trust in the process.

Operations

Production

GlassBrain Free Version's Beta, without any learning modules, is roughly 75% complete. It has been solely developed by Cade Harger, the founder up to this point. Once the free edition has been created, GlassBrain will begin creating a 'module reader' that will load in files that form modules. This will make the creation of educational modules efficient and something that anybody can do if GlassBrain ever were to decide to outsource module creation. GlassBrain will create the modules itself and then plans to have the content reviewed and critiqued by experts. This is significantly cheaper than having experts create the content themselves while maintaining the benefits that would come from doing so and will make the modules feel more centralized. All AWS setup and API call implementation will be outsourced.

Product Distribution

GlassBrain will be a downloadable desktop application that will be distributed through the GlassBrain website. The GlassBrain website will use Squarespace combined with Stripe to collect payments. Once payment is received, a product key will be given, which can be redeemed in the app to receive either the Learning Module or Premium Version that was purchased. Because of this, GlassBrain will be kept closed-source and sold on a per-computer basis to avoid reverse engineering of product keys.

Further down the line, GlassBrain will build its own custom site, with accounts to manage products so it doesn't have to be on a per-computer basis and so keys cannot be reverse-engineered. This site will also pave the way for further projects, such as a community workplace where people can share models.

The desktop app uses a React.js frontend with a Python/Flask backend and uses Electron.js to convert the web app into a desktop application. This allows for the potential to expand GlassBrain to be a web app as well in the future while maintaining only one primary codebase.

Why a Desktop App?

A desktop application is ideal as opposed to a web app as it runs everything on the end user's computer. This is beneficial because:

- It keeps development simple, not having to manage servers.
- It keeps pricing accessible and pricing models simple if cloud computing isn't used.
- Security concerns are almost zero because nothing is connected to the internet besides user feedback, which isn't sensitive and is secured through AWS.

Machine Learning is computationally expensive. While GlassBrain is fully capable of training large machine learning modules, this is a time-consuming task that doesn't help the learning process. During earlier learning modules, users will train models on their computers. The models in earlier modules will be smaller and quicker to train. In later modules, they will build models but then use pre-trained versions to cut the waiting out and save computation.

Other Logistics:

- Datasets will be sourced from publicly available datasets that can be downloaded from externally hosted sources. This keeps GlassBrain's download size smaller.
- Amazon Web Services (AWS) feedback collection will come in the form of API calls to store user feedback and feature usage stats in an AWS database. This is performed every time the user completes a module, every time there is a bug, and on a timer (perhaps every 30 minutes, depending on cost). After describing this to someone experienced in setting up AWS, the cost is likely to be less than 10 cents over the lifetime of a user.

Production Costs/Margins

GlassBrain Free Version (\$0):

- One-time costs:
 - Development costs are next to zero as all development is done in-house and unsalaried. Occasionally freelance work for graphic design/other things will add up.
- Recurring costs:
 - Maintaining a website - \$27/month

- AWS feedback collection - $\$<0.01/\text{Feedback Request}$
- Margin: $\sim \$-0.10/\text{user}$

GlassBrain Learning Modules (\$2-10):

- One-time costs:
 - Module production - \$100: Allocating a small budget to pay for the creation of custom graphics
 - Expert reviewing - \$400: Paying 2-3 experts to go through the module and give feedback, critique, and potentially use their name to achieve 'expert-reviewed' status
- Recurring costs:
 - AWS feedback collection - $\$<0.01/\text{Feedback Request}$
 - Transaction fees: 2.9% + 30¢
- Margin: $\sim 90.4\%$

GlassBrain Premium (\$50):

- One-time costs:
 - Same as the free version
- Recurring:
 - AWS feedback collection - $\$<0.01/\text{Feedback Request}$
 - Transaction fees: 2.9% + 30¢
- Margin: $\sim 96.3\%$

Operational Complexity

Maintaining GlassBrain will be simple and low-effort. Tasks to perform will consist of:

- Carrying out marketing (Posting on forums, creating ads occasionally)
- Monitoring sales (Checking KPIs, CRM tool, adjusting website accordingly)
- Carrying out customer support (checking email, monitoring feedback)
- Managing public image through social media
- With any spare time, working on new features or learning modules

Once the marketing and sales-oriented tasks ramp up (summer 2024), GlassBrain will hire at least one more person to help manage the workload and put non-technical things in more qualified hands. Later when GlassBrain scales, managing partnerships and selling to institutions will result in a dramatic increase in complexity, so at this point, more people will

need to be hired to help, or existing employees will need to increase commitment. More on these roles is in the [Management section](#).

Resources

Since GlassBrain is entirely software, no materials are required. Even if eventually GlassBrain merchandise is created, that will all be outsourced on Redbubble.

Financial Viability

See the next page for a conservative projection of GlassBrain sales/revenue for the years 2024 and 2025. (Click [here](#) to make a copy and experiment with numbers)

	Free to Premium Conversion Rate:	0.15	Lifetime server cost per user:	\$0.10	Notes:	With a soft launch in May, marketing will quickly ramp up over the summer. During the school year pay will be less because our team will be in school not working full time like the summer/christmas break. Over time, marketing money will become more effective as well.								
	Average # of modules purchased by free users:	0.8												
GlassBrain LLC														
Financial Statement														
For the Year Ending December 31st, 2024														
	January	February	March	April	May	June	July	August	September	October	November	December	YTD	
Revenue														
GlassBrain New Users					100	250	500	1000	1500	1500	1500	2000		
GlassBrain Premium Sales					15	37.5	75	150	225	225	225	300		
GlassBrain Premium Revenue					\$750	\$1,875	\$3,750	\$7,500	\$11,250	\$11,250	\$11,250	\$15,000	\$62,625	
Learning Module Revenue					\$408	\$1,020	\$2,040	\$4,080	\$6,120	\$6,120	\$6,120	\$8,160	\$34,068	
Other revenue	0	0	0	0	0	0	0	0	0	0	0	0	0	
Net Sales	\$0	\$0	\$0	\$0	\$1,158	\$2,895	\$5,790	\$11,580	\$17,370	\$17,370	\$17,370	\$23,160	\$96,693	
Cost of Goods Sold					-\$10	-\$25	-\$50	-\$100	-\$150	-\$150	-\$150	-\$200	-\$835	
Processing Fees (2.9% + \$0.3)	\$0.00	\$0.00	\$0.00	\$0.00	-\$38.08	-\$95.21	-\$190.41	-\$380.82	-\$571.23	-\$571.23	-\$571.23	-\$761.64	-\$3,179.85	
Gross Profit	\$0	\$0	\$0	\$0	\$1,168	\$2,920	\$5,840	\$11,680	\$17,520	\$17,520	\$17,520	\$23,360	\$92,678.15	
Operating Expenses														
Salaries & Wages	\$0	\$0	\$0	\$0	\$0	\$5,000	\$5,000	\$5,000	\$2,000	\$2,000	\$2,000	\$4,000	\$25,000	
Freelance expenses	\$0	\$0	\$2,000	\$2,000	\$2,000	\$2,000	\$2,000	\$2,000	\$2,000	\$2,000	\$2,000	\$2,000	\$20,000	
Depreciation Expenses	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	
Office Expenses	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	
Travel Expenses	\$0	\$50	\$0	\$50	\$0	\$50	\$0	\$50	\$0	\$50	\$0	\$50	\$300	
Subscriptions/upkeep	\$50	\$50	\$50	\$50	\$100	\$100	\$100	\$100	\$100	\$100	\$100	\$100	\$1,000	
Advertising Expenses	\$0	\$0	\$0	\$0	\$500	\$1,500	\$3,000	\$3,000	\$1,000	\$1,000	\$1,000	\$4,000	\$15,000	
Total Operating Expenses	\$50	\$100	\$2,050	\$2,100	\$2,600	\$8,650	\$10,100	\$10,150	\$5,100	\$5,150	\$5,100	\$10,150	\$61,300	
Income Before Income Taxes	-\$50	-\$100	-\$2,050	-\$2,100	-\$1,432	-\$5,730	-\$4,260	\$1,530	\$12,420	\$12,370	\$12,420	\$13,210	\$31,378	
Income Tax Expense								-\$234	-\$1,900	-\$1,893	-\$1,900	-\$2,021	-\$4,801	
Net Income	-\$50	-\$100	-\$2,050	-\$2,100	-\$1,432	-\$5,730	-\$4,260	\$1,296	\$10,520	\$10,477	\$10,520	\$11,189	\$26,577	
Balance:	25,000	\$24,950	\$24,850	\$22,800	\$20,700	\$19,268	\$13,538	\$9,278	\$10,574	\$21,094	\$31,571	\$42,091	\$53,280	

	Free to Premium Conversion Rate:	0.2	Lifetime server cost per user:	\$0.10	Notes:	I expect 2,500 new users every month to be a sustainable target. In December of 2024, GlassBrain will begin selling contracts to schools. In May, pay will increase to support multiple people in full-time jobs.							
	Average # of modules purchased by free users:	0.8											
GlassBrain LLC													
Financial Statement													
For the Year Ending December 31st, 2025													
	January	February	March	April	May	June	July	August	September	October	November	December	YTD
Revenue													
GlassBrain New Users	2500	2500	2500	2500	2500	2500	2500	2500	2500	2500	2500	2500	
GlassBrain Premium Sales	500	500	500	500	500	500	500	500	500	500	500	500	
GlassBrain Premium Revenue	\$25,000	\$25,000	\$25,000	\$25,000	\$25,000	\$25,000	\$25,000	\$25,000	\$25,000	\$25,000	\$25,000	\$25,000	\$300,000
Learning Module Revenue	\$9,600	\$9,600	\$9,600	\$9,600	\$9,600	\$9,600	\$9,600	\$9,600	\$9,600	\$9,600	\$9,600	\$9,600	\$115,200
School contracts	\$0	\$1,000	\$0	\$3,000	\$3,000	\$0	\$3,000	\$0	\$3,000	\$3,000	\$3,000	\$0	\$19,000
Net Sales	\$34,600	\$35,600	\$34,600	\$37,600	\$37,600	\$34,600	\$37,600	\$34,600	\$37,600	\$37,600	\$37,600	\$34,600	\$434,200
Cost of Goods Sold	-\$250	-\$250	-\$250	-\$250	-\$250	-\$250	-\$250	-\$250	-\$250	-\$250	-\$250	-\$250	-\$3,000
Processing Fees (2.9% + \$0.3)	-\$1,153.40	-\$1,182.40	-\$1,153.40	-\$1,240.40	-\$1,240.40	-\$1,153.40	-\$1,240.40	-\$1,153.40	-\$1,240.40	-\$1,240.40	-\$1,240.40	-\$1,153.40	-\$14,391.80
Gross Profit	\$34,850	\$35,850	\$34,850	\$37,850	\$37,850	\$34,850	\$37,850	\$34,850	\$37,850	\$37,850	\$37,850	\$34,850	\$416,808.20
Operating Expenses													
Salaries & Wages	\$5,000	\$2,000	\$2,000	\$2,000	\$2,000	\$25,000	\$25,000	\$25,000	\$25,000	\$25,000	\$25,000	\$25,000	\$188,000
Freelance expenses	\$2,000	\$2,000	\$2,000	\$2,000	\$2,000	\$2,000	\$2,000	\$2,000	\$2,000	\$2,000	\$2,000	\$2,000	\$24,000
Depreciation Expenses	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0
Office Expenses	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0
Travel Expenses	\$100	\$100	\$100	\$100	\$100	\$100	\$100	\$100	\$100	\$100	\$100	\$100	\$1,200
Subscriptions/upkeep	\$100	\$100	\$100	\$100	\$100	\$200	\$200	\$200	\$200	\$200	\$200	\$200	\$1,900
Advertising Expenses	\$5,000	\$5,000	\$5,000	\$5,000	\$5,000	\$5,000	\$5,000	\$5,000	\$5,000	\$5,000	\$5,000	\$5,000	\$60,000
Total Operating Expenses	\$12,200	\$9,200	\$9,200	\$9,200	\$9,200	\$32,300	\$32,300	\$32,300	\$32,300	\$32,300	\$32,300	\$32,300	\$275,100
Income Before Income Taxes	\$22,650	\$26,650	\$25,650	\$28,650	\$28,650	\$2,550	\$5,550	\$2,550	\$5,550	\$5,550	\$5,550	\$2,550	\$141,708
Income Tax Expense	-\$3,465	-\$4,077	-\$3,924	-\$4,383	-\$4,383	-\$390	-\$849	-\$390	-\$849	-\$849	-\$849	-\$390	-\$21,681
Net Income	\$19,185	\$22,573	\$21,726	\$24,267	\$24,267	\$2,160	\$4,701	\$2,160	\$4,701	\$4,701	\$4,701	\$2,160	\$120,027
Balance:	\$3,280	\$95,037	\$116,762	\$141,029	\$165,295	\$167,455	\$172,156	\$174,316	\$179,017	\$183,718	\$188,419	\$190,578	\$190,578

Financial Viability

As shown in the financial projections, GlassBrain will be able to sustainably profit while keeping a few employees full-time if it can sustainably bring on an average of only 2,500 new users (total, not just premium) every month. GlassBrain's extremely low recurring input costs help this number be so low. This number is also after thousands are spent each month on marketing. Given the course enrollment numbers of Coursera and Udemy mentioned in the [Market and Strategy section](#), this seems like an easily achievable goal.

GlassBrain is to win the Stumberg Final Round Competition, the \$25,000 in prize money will be sufficient to jumpstart the business so no further funding will be sought out. Potentially in the long term, more funding could be in the picture, but only if GlassBrain is planning a big expansion, such as creating a professional version of the software.

Management

Team (solo)

Founder, Cade Harger | Junior at Trinity University, Major in Computer Science, GPA: 3.68 / 4.0
Cade is responsible for everything currently. Once more employees are brought on, responsible for overseeing all operations, making executive decisions, and creating features on the Desktop App. He will also be responsible for creating initial versions of Educational Modules, which will then be reviewed by experts.

Relevant Skills:

- Programming experience in 10 languages and 2 Machine learning frameworks
- Solely created GlassBrain Desktop App until the current point in time:
 - Used Javascript, React.js, Redux.js, and React Flow to create the frontend.
 - Utilized Python and Flask to compile the frontend representation to Pytorch.
 - The most complex parts are complete, and the proof of concept (building-block interface) is running.
- Completed the Geekdom Summer Incubator program on top of the Stumberg Summer Accelerator, building a better network and collecting feedback from experienced people in the process.
- Lead classes teaching Scratch (the precedent for GlassBrain, mentioned in the [Market and Strategy section](#)) to children ages 5-14 in a summer camp
- Solely programmed, designed, and released an iPhone app that allowed users to create a simulated stock market portfolio and buy or sell virtual stocks based on the real stock market. Updated the app to address issues.
- Created many machine learning personal projects, such as a chess-playing program and novel hybrid neural network architectures.
- Developed project management skills while becoming an Eagle Scout
- Self, school, and online course-taught in machine learning and has experienced the issues that GlassBrain was created to solve firsthand and seen his peers struggle with them too in school.
- View a record of work done during the Summer Accelerator in the [Appendix](#).

Future Team Members

GlassBrain will offer new team members who join to work over the summer of 2024 equity (up to 30% across all new team members) as well as pay. Team roles are mentioned at the end of the Operations section. Team members would ideally be given a decent degree of freedom, as they would be more knowledgeable in their respective fields, but final decisions will need company-wide approval. Here are the job flyers for the two positions GlassBrain will ideally hire if the Final Round Prize is won:

GlassBrain STARTUP HIRING - Media/Marketing Lead:

Description:

Starting in the summer of 2024, this role will be responsible for managing the marketing side of GlassBrain and carrying out marketing campaigns.

Key responsibilities:

- Devise marketing strategies and manage small budget to generate leads
- Create advertisements, social media posts
- Meet and discuss with the founder and potentially other team members

Qualities:

- Willing to become passionate about the company
- Creative, spots opportunities for guerilla marketing and new channels
- Some education in marketing
- Ideally previous experience with marketing campaigns and/or social media
- Willing to work at least 10 hours per week during the school year

About GlassBrain:

Winner of the 2023 Stumberg \$5,000 seed-round entrepreneurship competition, GlassBrain is a desktop app that teaches individuals AI concepts in a more visual, interactive, and intuitive way. Currently, the app is in development.

Pay:

Negotiable. This will depend on the legal status of your employment, whether as an independent contractor, employee, or equity partner. You can expect somewhere in the range of \$9-14 an hour.

Interested? Contact:

Cade Harger:

Phone: (713)-438-5084 Email: glassbrainai@gmail.com Instagram: @glassbrainai

GlassBrain STARTUP HIRING - Sales, UX Lead:

Description:

Starting in the summer of 2024, this role will be responsible for managing the user experience side of GlassBrain and converting leads into paying customers.

Key responsibilities:

- Create mockups and advise developer(s) on product/website user experience
- Track provided KPIs and devise explainable strategies to increase them
- Be the first point of contact for potential customers
- Meet and discuss with the founder and potentially other team members

Qualities:

- Willing to become passionate about the company
- Personable, persistent, and able to understand the motivations of the customer
- Some education in sales, previous experience is a plus
- Previous experience or provable talent in UX
- Willing to work at least 10 hours per week during the school year

About GlassBrain:

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Interested? Contact:

Cade Harger:

Phone: (713)-438-5084 Email: glassbrainai@gmail.com Instagram: @glassbrainai

Mentorship

GlassBrain has many mentors who are eager to help.

At Trinity University: The faculty and staff are extremely supportive and willing to help in any way they can. This includes not just the Entrepreneurship department, but also all others.

At [Trinity Venture Mentoring Services \(VMS\)](#): GlassBrain applied and received two mentors: [Corky Roth](#) and [Derick Rodgers](#), both with extensive experience working with/helping startups. GlassBrain has recurring meetings (two so far) with these mentors to discuss progress and issues the venture is facing.

At [Geekdom](#): During GlassBrain's participation in the Summer Geekdom Incubator, connections were fostered and GlassBrain got helpful feedback at every meeting. GlassBrain

intends to purchase a Geekdom Student membership once the development side of the desktop app is complete to begin gaining advice on business-related things. Geekdom has many mentors in many different fields and a membership to gain access to them is invaluable.

Use of Funding

Use of Seed Round Funds (\$5,000)

Item	Date	Balance Change	Status
INITIAL BALANCE	07/24	\$5,100.00	Recieved
GlassBrain LLC Founding	09/26	-\$300.00	Paid
Initial Bank deposit	09/26	-\$100.00	Paid
Trademark Filing	07/31/23 and 08/08/23	-\$697.00	Paid
Gas/Parking	09/26	-\$100.00	Paid
Monthly Bank fee	09/01/23	-\$16.00	Paid
Freelance logo design	09/26	-\$68.80	Paid
Result Balance:		\$3,818.20	

Given the current stage of GlassBrain's development, it hasn't made sense to spend very much. Since GlassBrain isn't in business yet, there are no costs associated with marketing or sales. As for spending to increase app development speed, this is something that GlassBrain will heavily consider after its initial release. This would work well for things like expansions, optimization, and quality-of-life changes. It won't be completely feasible until then because:

- **The app is visionary:** When communicating what work needs to be done, it wouldn't be as simple as saying "build a module builder", as nothing is existing to go off of. If features aren't implemented nearly perfectly, which is likely, they would need to be redone as otherwise it would defeat the purpose of GlassBrain.
- **The app is incomplete as a standalone project:** This means that any features that would be outsourced would need to be built from scratch, worsening the vision issue.
- **The code is complex:** Given the nature of machine learning concepts and the creation of a compiler, the labor for some parts of the app would need to be outsourced to someone fairly experienced, which would likely cost more than the budget the Seed Round funding allows for.

Banking is done through [Bank Of America Business Advantage](#).

Planned Use of Final Round Funds (\$25,000)

\$10,000 (%) | Creation of Education Content:

This amount will be used to get experts to review the initial versions of Education Modules and make edit suggestions. Experts might include professors, researchers, and people who work in desirable jobs in the machine learning field. This gives the modules credibility and gives GlassBrain the ability to advertise ‘Expert-Reviewed, PHD approved’ status.

Having the initial versions of the modules created in-house has many benefits. It allows for the cheap creation of Educational Modules because the experts aren’t creating the content themselves, only looking over it. If an additional feature or way of viewing content would improve the experience, it is within the scope of module creation to add the functionality to the desktop app. Also, GlassBrain has a more centralized feel if modules come from one source. The vision of GlassBrain and what its modules should do to solve pain points is preserved this way as well.

\$8,000 (%) | Marketing:

This will provide a significant budget to be able to jumpstart GlassBrain’s growth as described in the [Market and Strategy section](#).

\$5,000 (%) + Remaining seed round funds | Freelance/paying future employees

There will inevitably be work outside of the skill set of GlassBrain employees. This will be outsourced to freelance work through connections and freelance matching websites. This amount also allows for enough time to pay at least one employee until profitability is reached. Setting up AWS to collect user feedback will fall under this category.

\$1,000 (%) | Sales/website optimization:

This amount is allocated to pay for the GlassBrain website (\$27 + Domain fees/month), optimize the website, and optimize the site’s SEO to maximize ROI by raising click-through rates. This could also pay for a quality CRM tool.

\$1,000 (%) | Legal:

Through connections at [Geekdom](#), informal legal direction is virtually free. This will help GlassBrain figure out what needs to be done to ensure all laws are being complied with. The allocation of money will help GlassBrain draft documents such as terms and conditions from experts to minimize the risk of any lawsuit. It’s worth noting the threat of GlassBrain

legally infringing upon other companies is also minimized by GlassBrain being a unique technology.

Operating Agreements and Bylaws

As GlassBrain only has one employee, no Operating Agreements or Bylaws are needed at the moment. Additional Operating Agreements and Bylaws will be added once GlassBrain adds equity partners or employees. Trinity Entrepreneurship has provided resources to assist their eventual creation as a part of the Summer Accelerator Program and there are many resources on the internet as well.

Legally, GlassBrain LLC is a Texas LLC. Since there is only one employee, GlassBrain is treated legally similarly to a sole proprietorship (besides liability).

Prototype

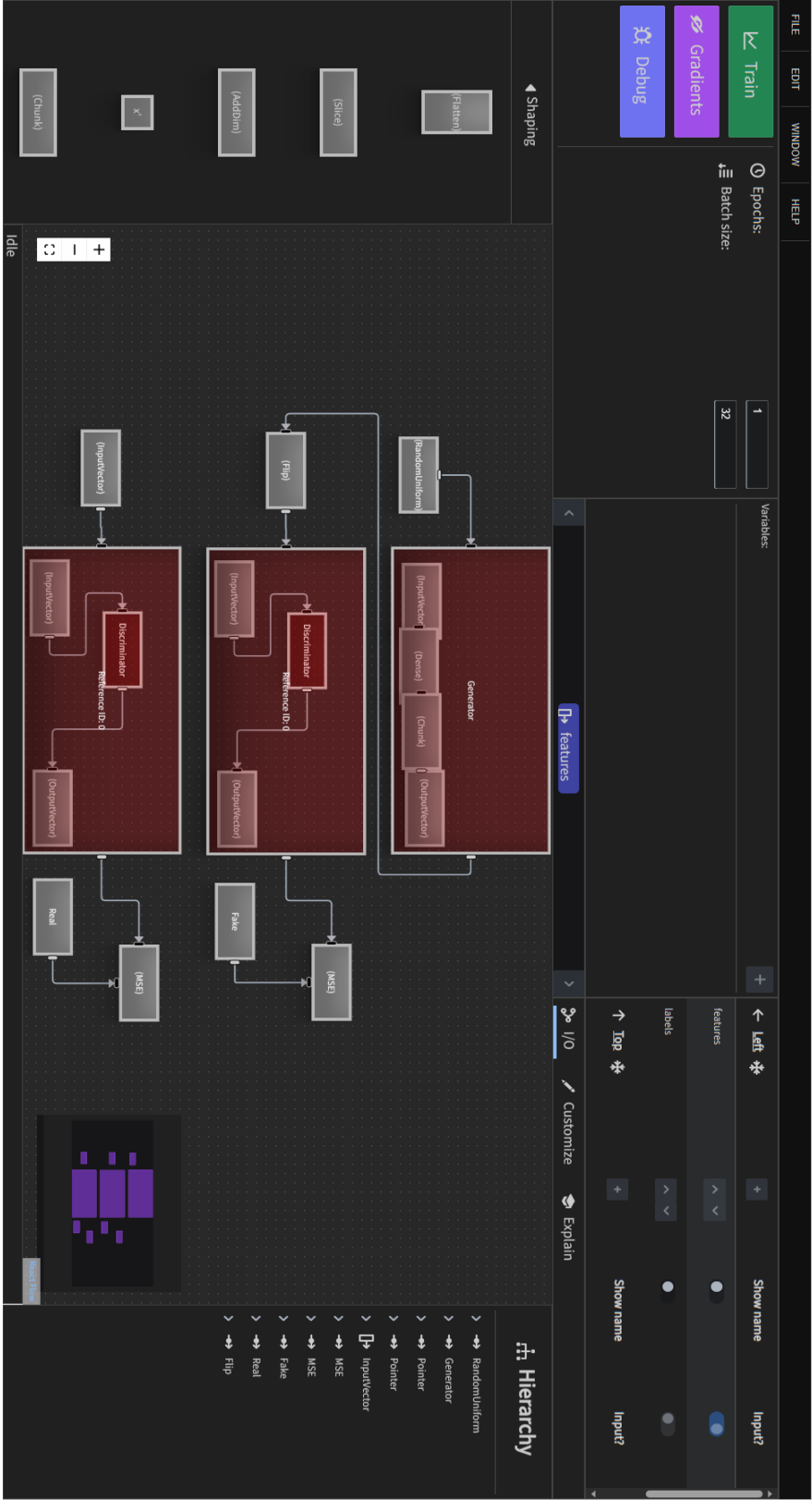
GlassBrain Free Version is roughly 75% complete. GlassBrain, with Learning Modules and Premium, is expected to fully release at the beginning of the summer of 2024.

Existing Functionality

- Support for MacOS, Windows, and Linux
- Graphically build and train almost any neural network architecture
- Representational freedom: The user can modify the appearance of architectures to a reasonable degree to make their projects more readable and universally understandable.

Functionality Needed Before Release

- The ability to load external datasets (Currently in progress): As of now, only the MNIST digit classification is built in and used for testing purposes.
- The ability to test and visualize the results of trained networks
- As of now, a few common architectures are built in: (Convolutional layers, Dense layers, Multi-head attention, LSTM, Etc.). Adding a few more of the most commonly used architectures would be beneficial
- Speed changes: Because GlassBrain compiles directly to Pytorch, after optimization, network speed should be comparable to Pytorch performance (industry standard). In-depth optimization changes have not yet been made, so some models run at 99% of Pytorch's speed while others run up to 75x slower.
- Error messages: when the user does something they shouldn't, GlassBrain needs to alert them.
- In-depth testing: There are still ways GlassBrain can crash. Testing will solve this.
- Saving/Loading projects
- GPU support?
- Creation of Learning Module Reader
- Creation of Learning Modules



An example of a simple Generative Adversarial Network (GAN) in GlassBrain

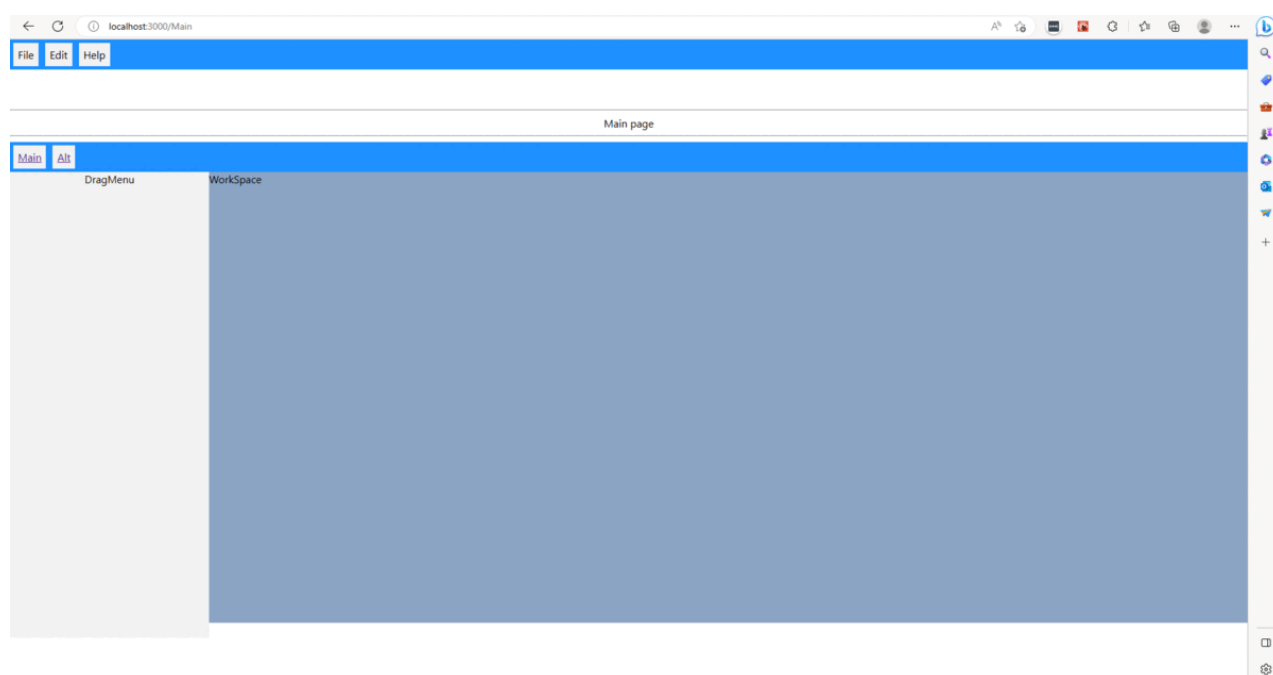
Here is a video of the creation of a simple neural network with 2 layers:

- https://youtu.be/i8r_3e3cEXw
- The input is passed into a model that flattens it, then runs it through two Dense layers, one with 50 neurons and the other with 10. It then applies Softmax activation and uses Categorical Cross Entropy to compare the output with the labels to provide gradients. The model is then trained to classify handwritten digits.
- At 1:58, it is visible from the terminal that the model has successfully trained.

Here are videos demonstrating some of GlassBrain's functionality:

- Part one: <https://youtu.be/EPF0r5B5t5w>
- Part two: <https://youtu.be/8tmvILCd7Ew>

Here is a video of screenshots of GlassBrain's progress over the summer:



Appendix

What are Neural Networks, and how are they different from Artificial Intelligence (AI) and Machine Learning (ML)?

- All of the headlines lately seem to mention AI. Technically speaking, machine learning is a subfield of AI, and Neural Networks are a subfield of machine learning. However, almost every single notable AI technology in recent years has been powered by machine learning, specifically neural networks.
- Because neural networks are responsible for so much of the field, the definitions have colloquially shifted. Nowadays AI is used to refer to the business or user front-end side of technologies, like ChatGPT. Machine learning is used to refer to the back-end, more scientific side of things, like the Transformer, which is the neural network architecture that powers ChatGPT. Neural networks are used to be more specific.
- All Large Language Models, self-driving cars, cancer-detection programs, Image generators like DALL-E 3, and FaceID to unlock your phone are all powered by neural networks.

[Here](#) is a copy of the founder's notes on work done each day during the Summer Accelerator.

Will GlassBrain become outdated as the field advances?

- No. While the field of AI is rapidly advancing, the concepts that we know to be valuable are neural networks, which is what GlassBrain teaches. You can think of it as the fact that despite ongoing research in mathematics, we still teach addition and subtraction to elementary school students because all advancements we make are based upon them.